Database Design

A simple SQLite database for student records

CampusDesk stores the essential details needed to identify and contact each student, record their programme and year of study, and keep each student in one row.

ENTITY-RELATIONSHIP DIAGRAM

STUDENTS

PK	id - integer, auto-increment	primary key
UQ	student_number - text, required	unique
	full_name - text, required	
UQ	email - text, required	unique
	phone - text, optional	
	programme - text, required	
	year_of_study - integer (1-8)	check
	created_at - timestamp, automatic	default

One student profile is stored per row; the project does not need join tables.

KEYS AND CONSTRAINTS

PK students.id uniquely identifies each record.

UQ student_number and email are required and unique.

CHECK year_of_study is restricted to integers from 1 through 8.

DEFAULT phone is optional; created_at is filled by SQLite.

SQLite Schema

The schema below creates the Students entity and its key and validation rules.

```
CREATE TABLE IF NOT EXISTS students (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    student_number TEXT NOT NULL UNIQUE,
    full_name TEXT NOT NULL,
    email TEXT NOT NULL UNIQUE,
    phone TEXT NOT NULL DEFAULT '',
    programme TEXT NOT NULL,
    year_of_study INTEGER NOT NULL
        CHECK (year_of_study BETWEEN 1 AND 8),
    created_at TEXT NOT NULL DEFAULT (datetime('now'))
);
```

APPLICATION RECORD OPERATIONS

ADD Validate form values and insert with a prepared SQLite statement.

VIEW List student records; search by number, name, email, or programme.

DELETE Confirm the action and delete the selected row by its primary key.

The application creates the SQLite database file in its data directory on first run.
The same creation statement is provided separately in schema.sql.